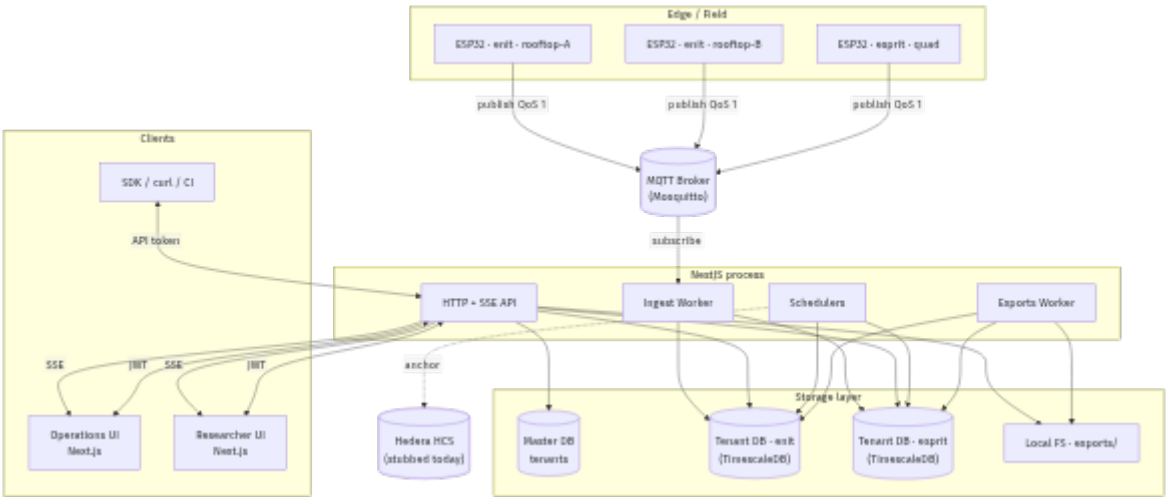


WeatherHub Backend – Architecture Diagrams

Pure-tech architecture reference. Diagrams only.

1. System topology



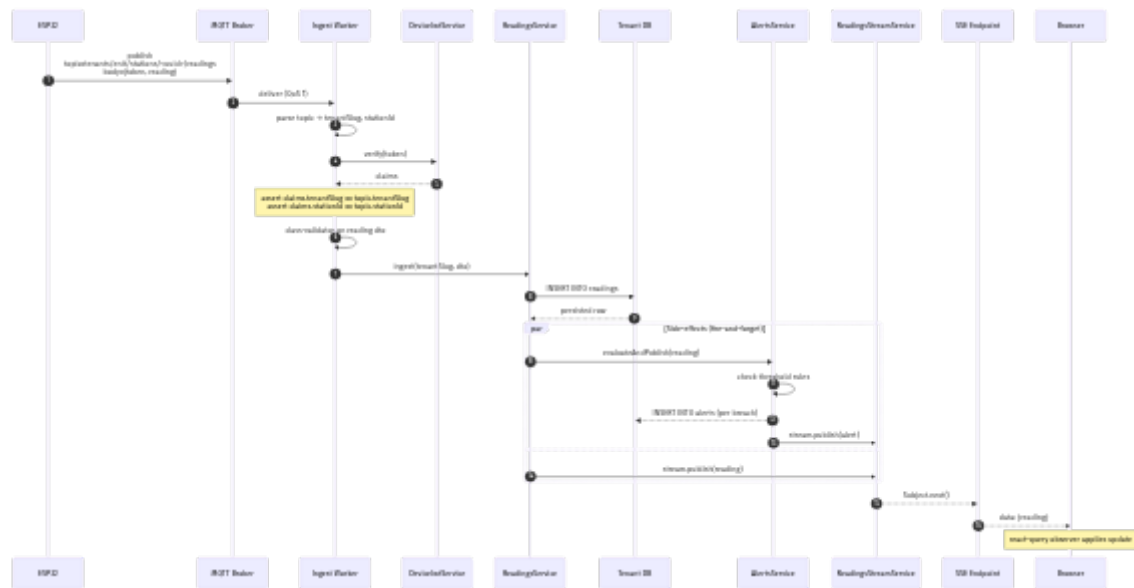
2. Multi-tenancy — database-per-tenant



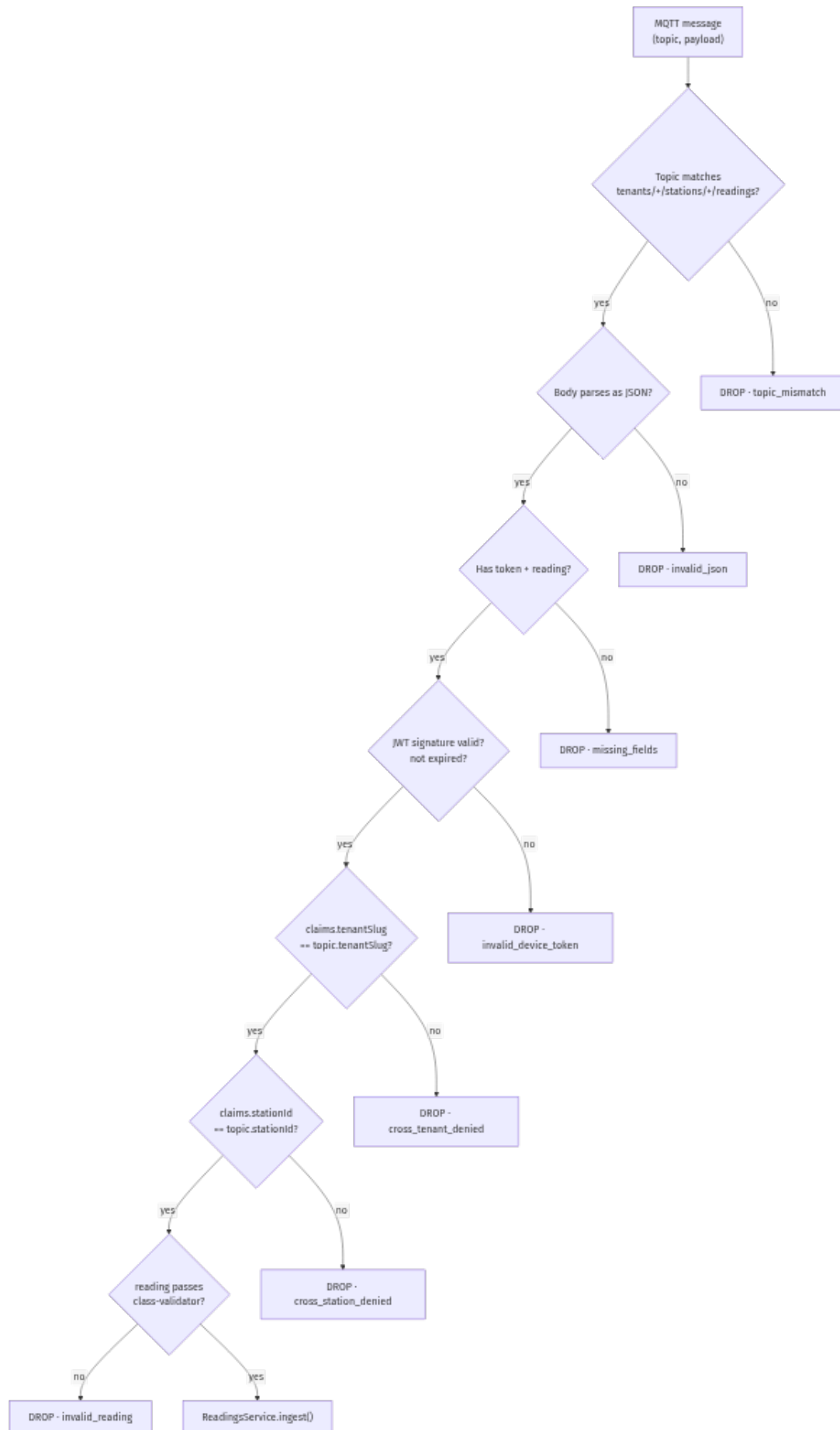
3. Per-tenant database schema



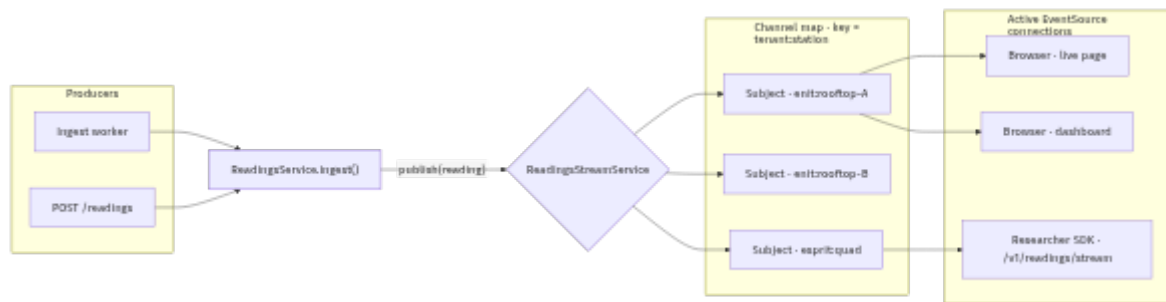
4. Live data path — ESP32 reading to browser



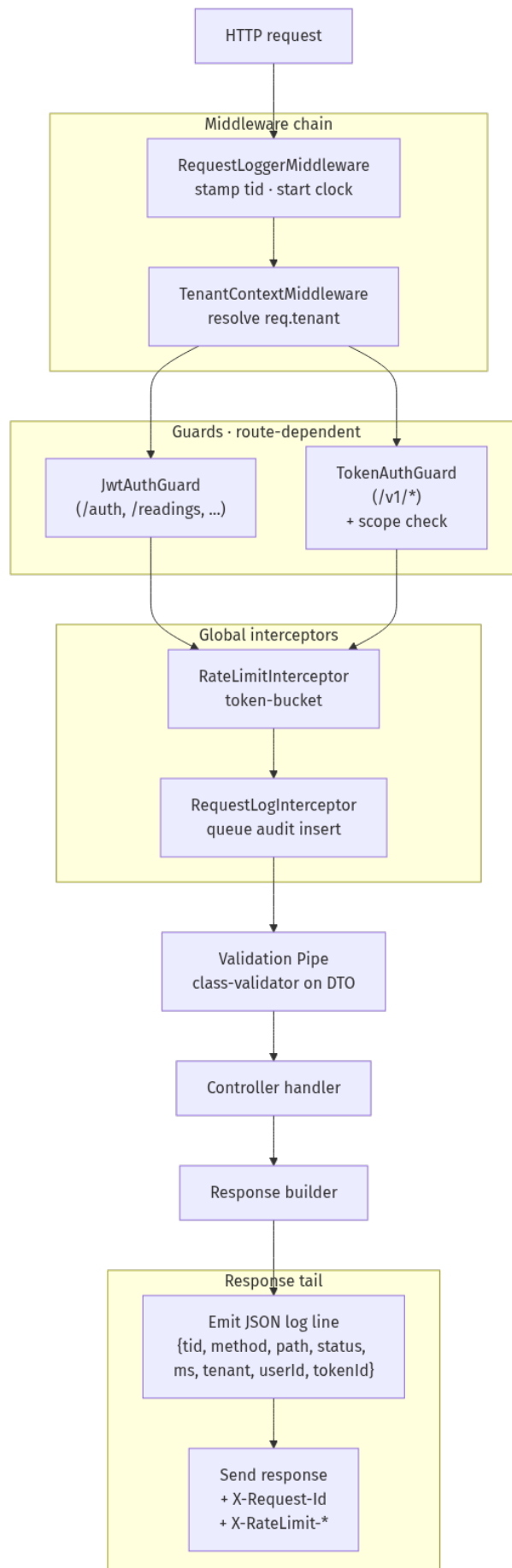
5. Ingest worker — message validation pipeline



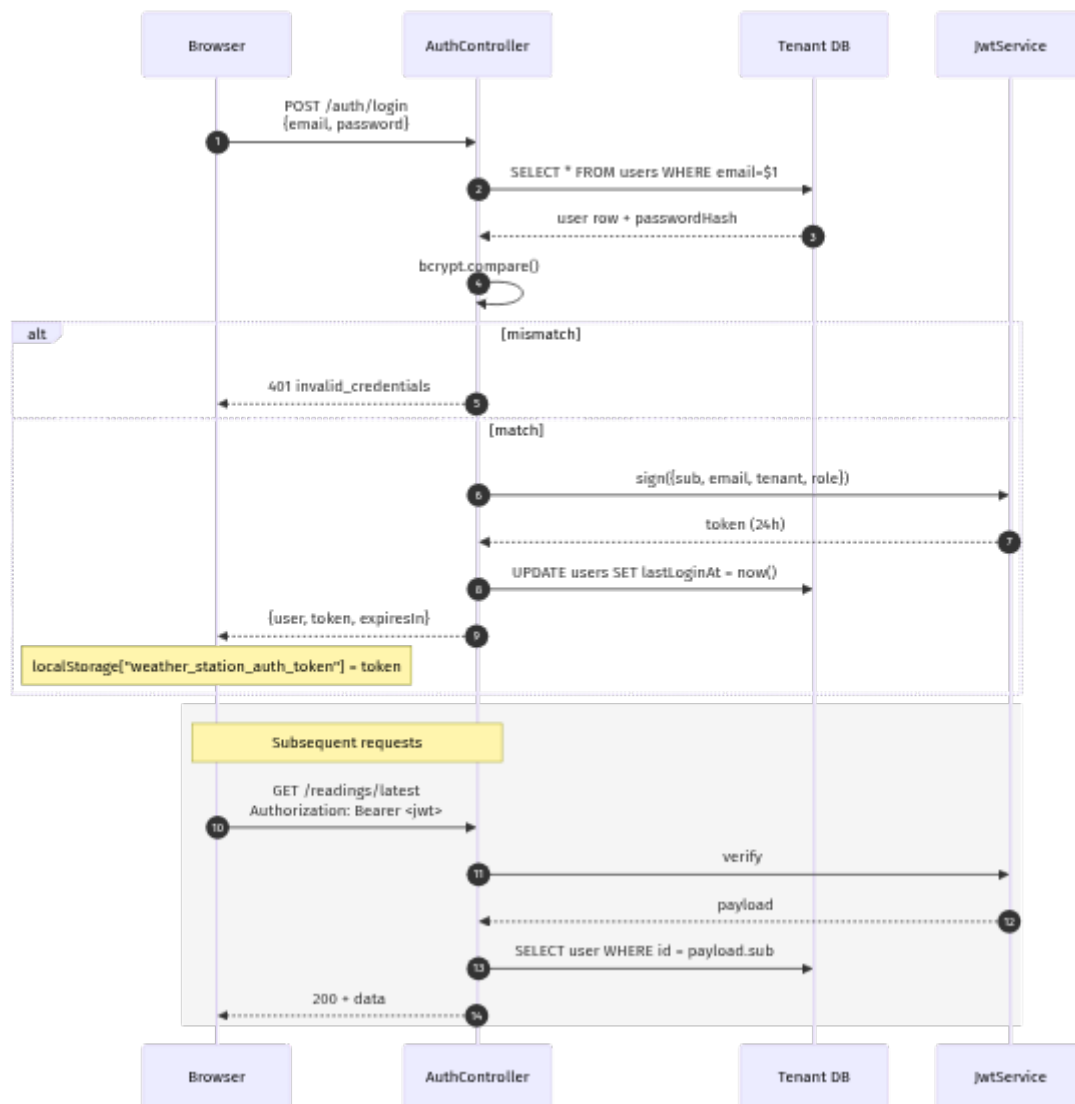
6. SSE fan-out



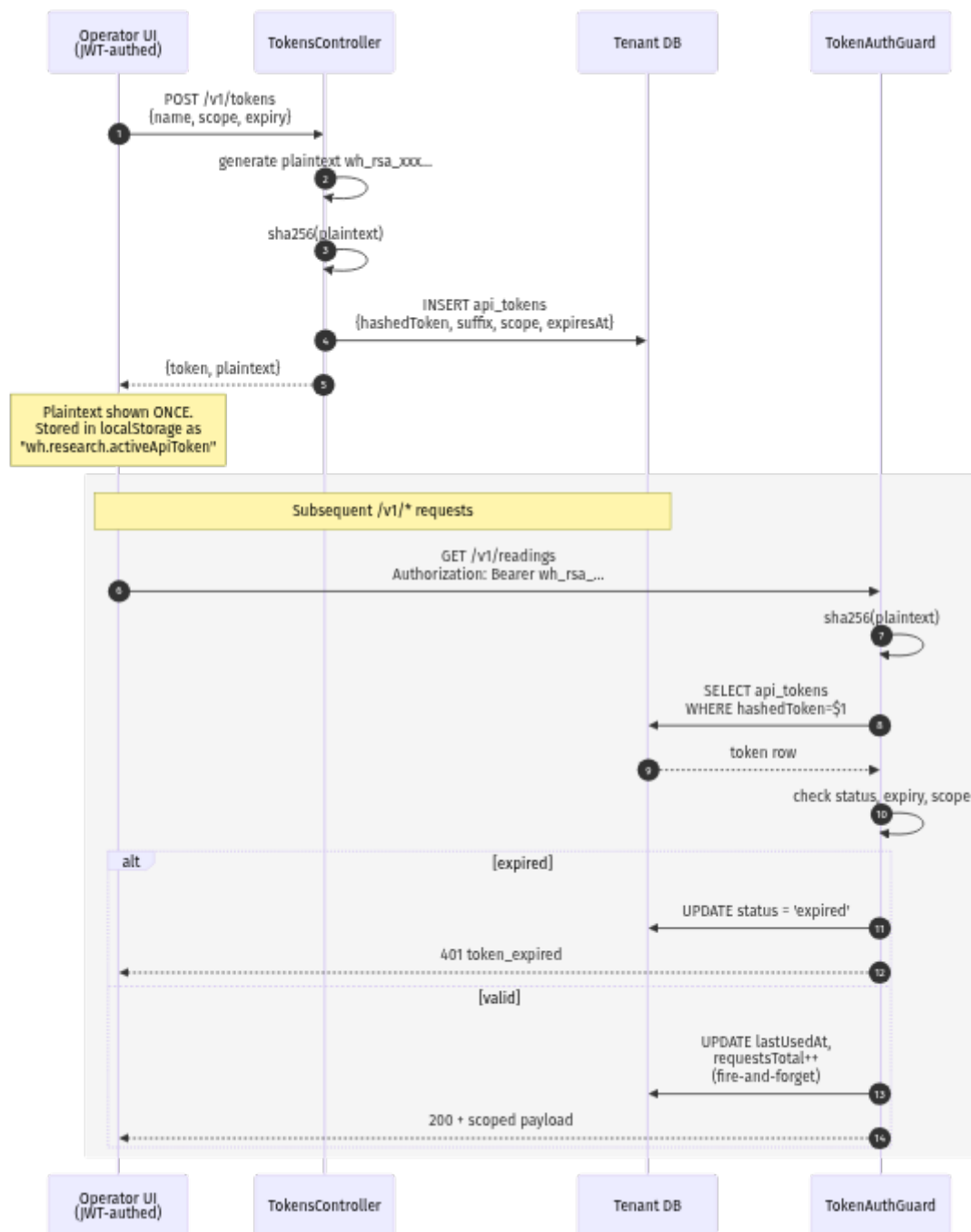
7. HTTP request lifecycle through NestJS



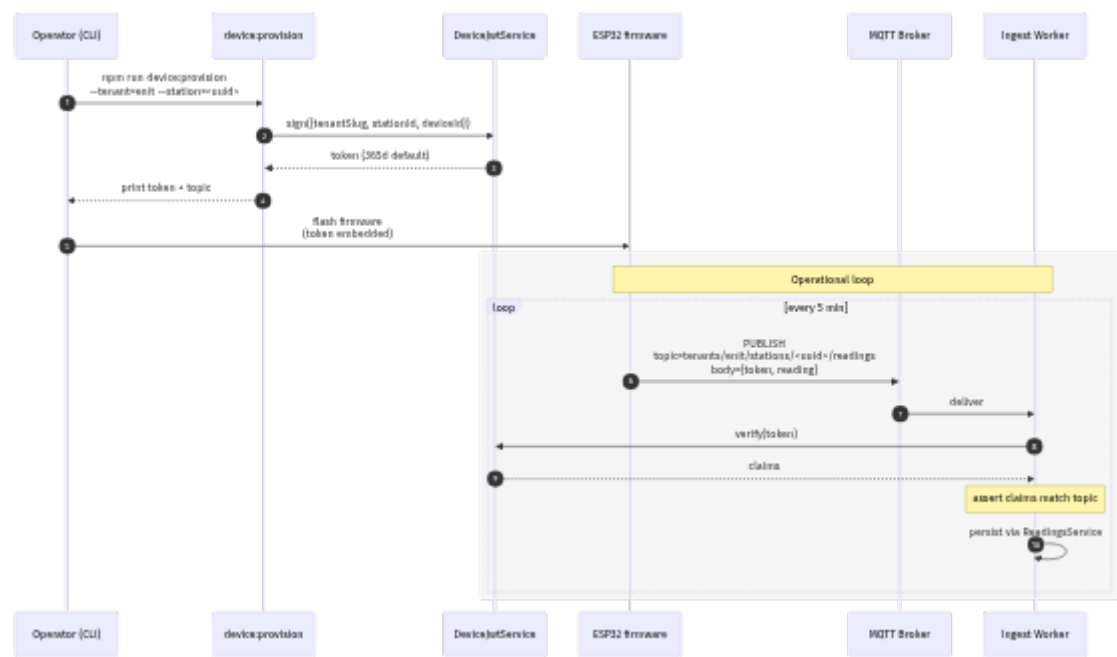
8. Auth flow A — Operator JWT session



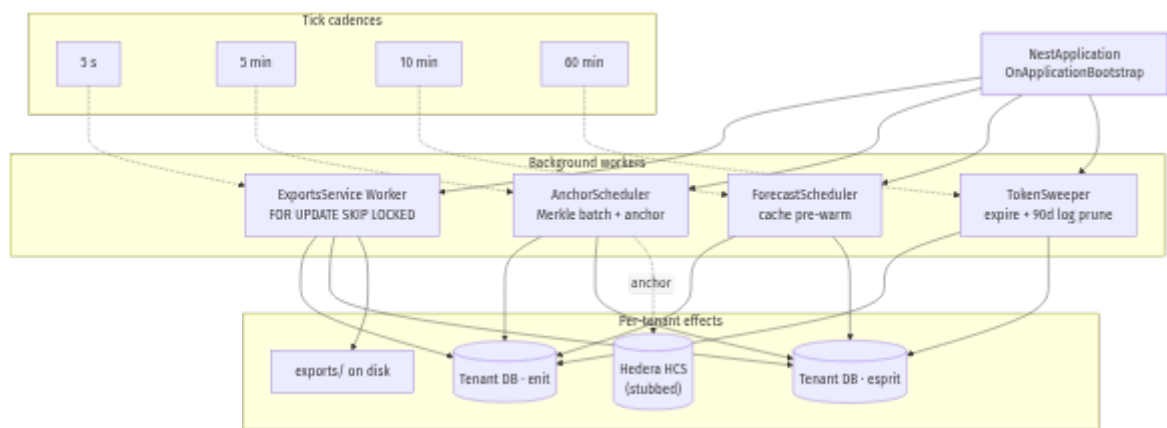
9. Auth flow B — API token (Researcher)



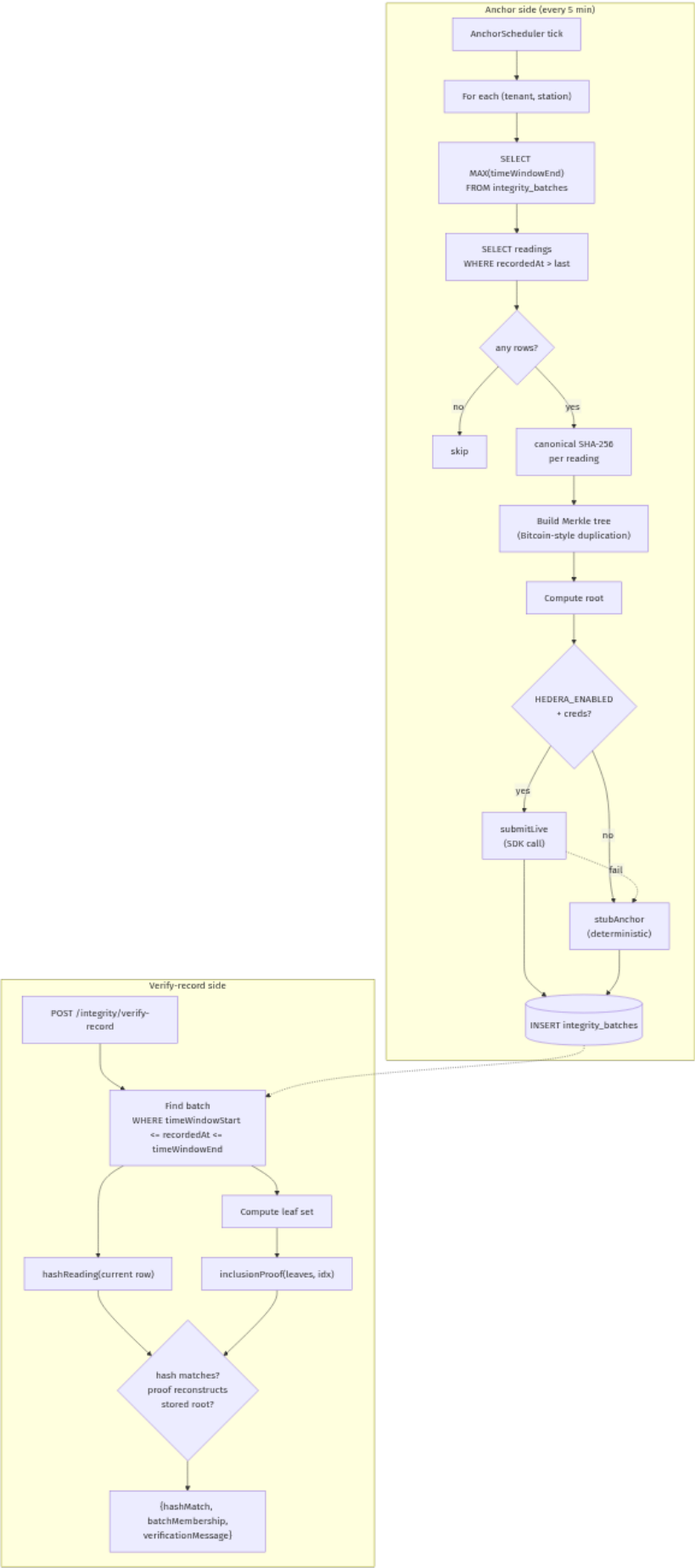
10. Auth flow C — Device JWT (MQTT only)



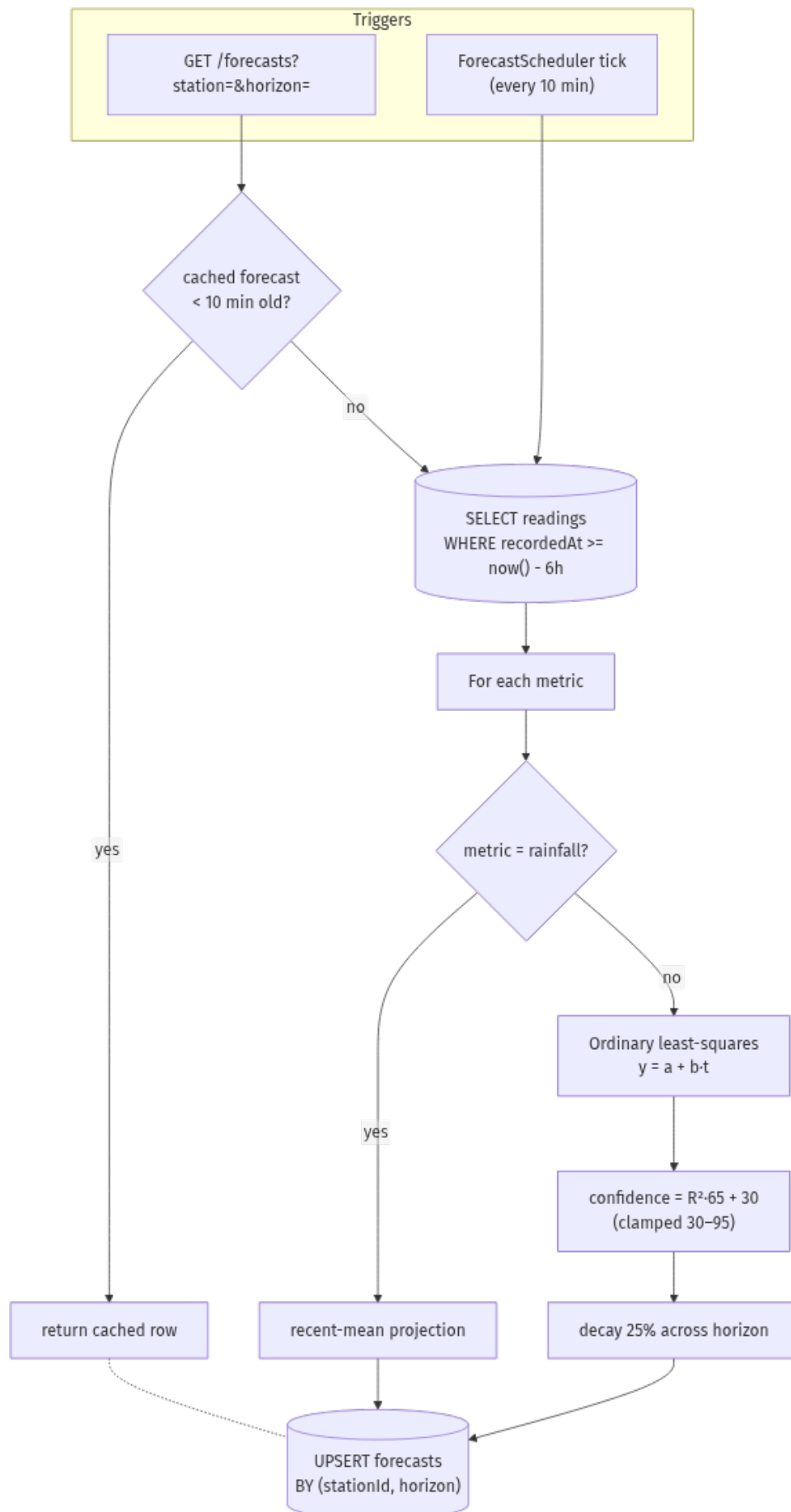
11. Background workers — schedulers + queues



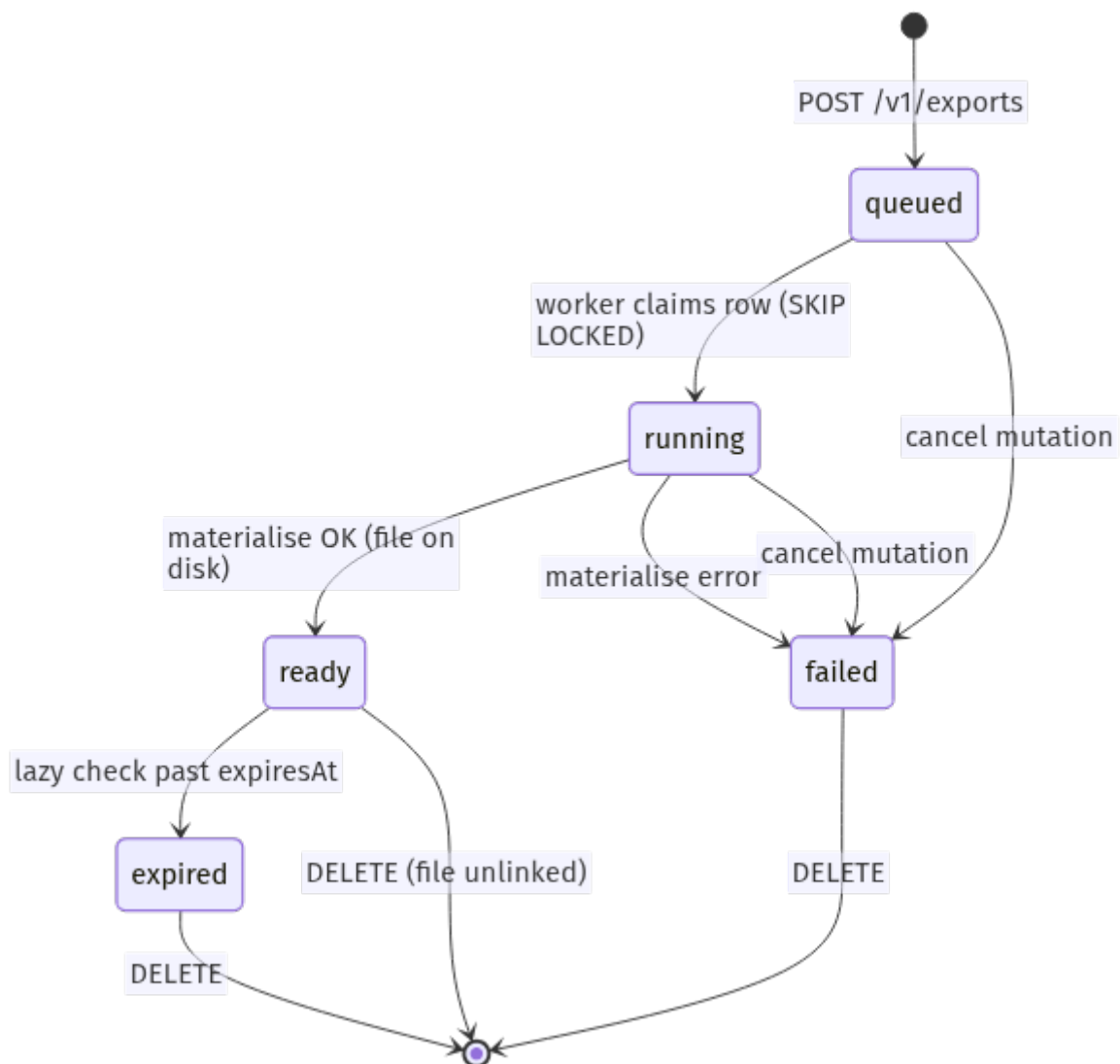
12. Anchor + integrity verification



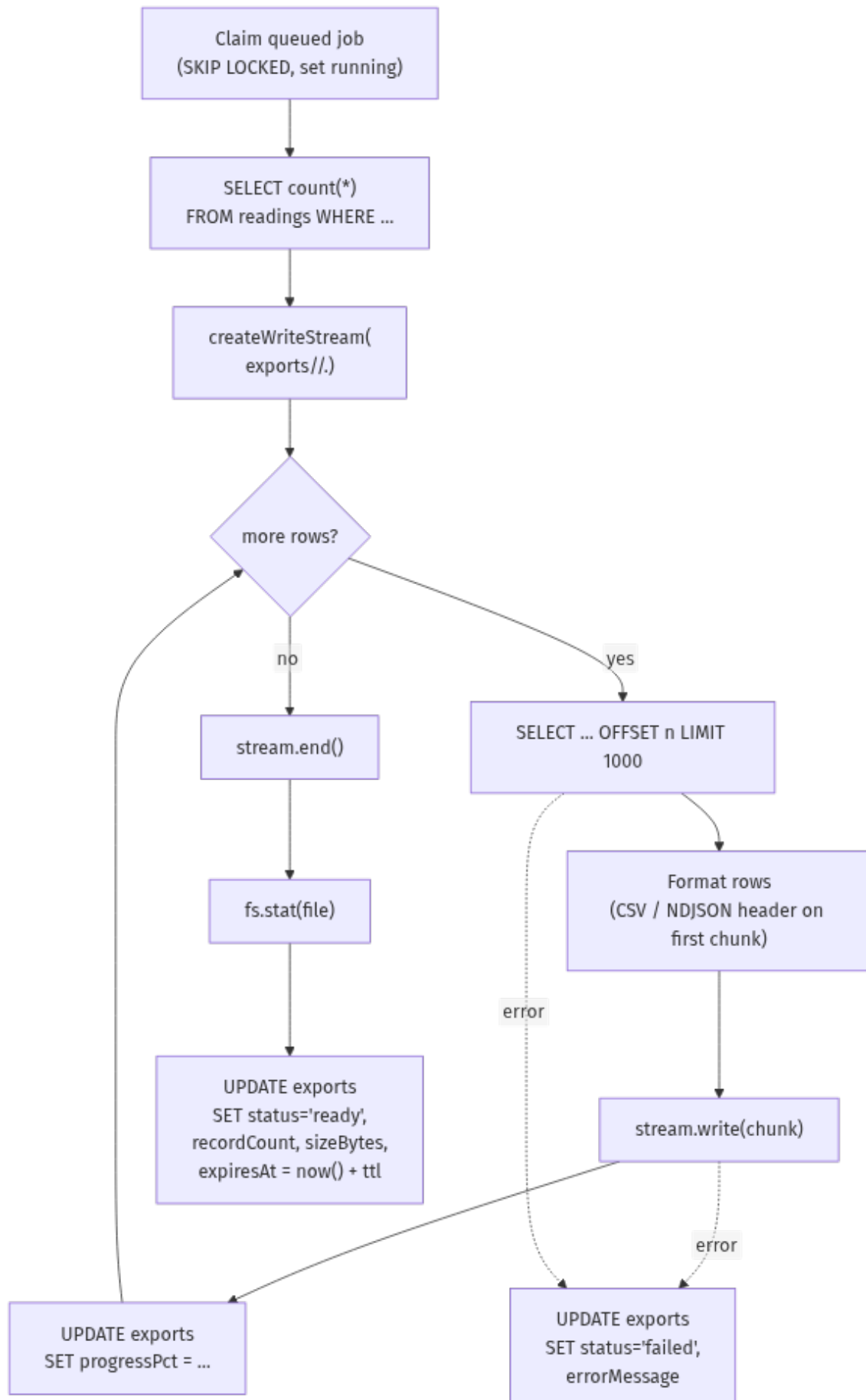
13. Forecast generation



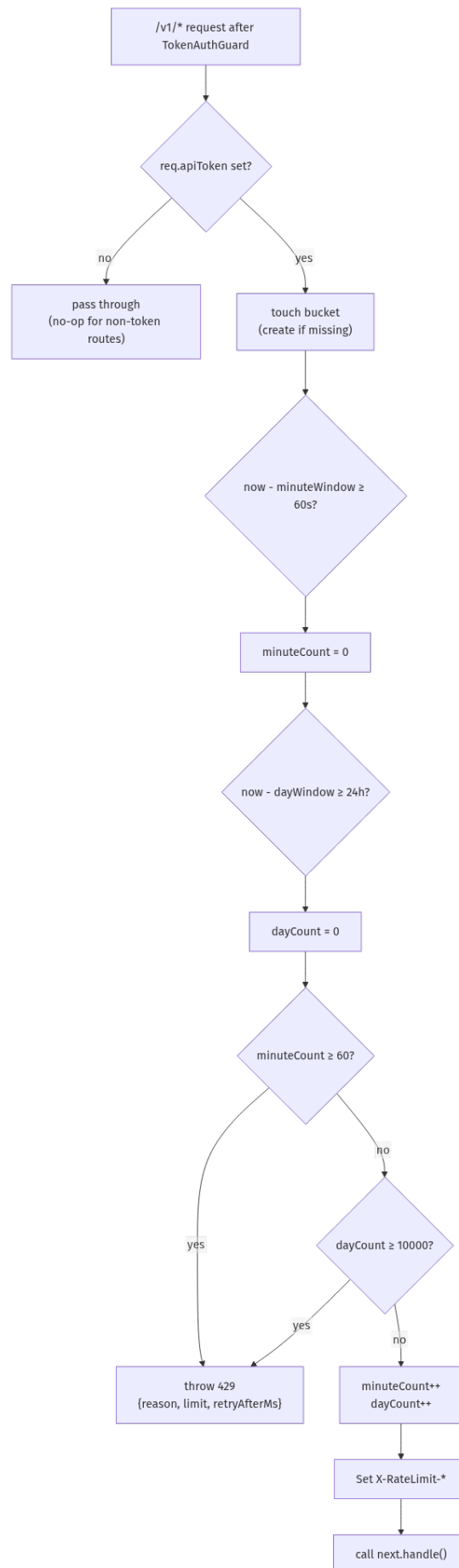
14. Export job state machine



15. Export materialiser — streaming write



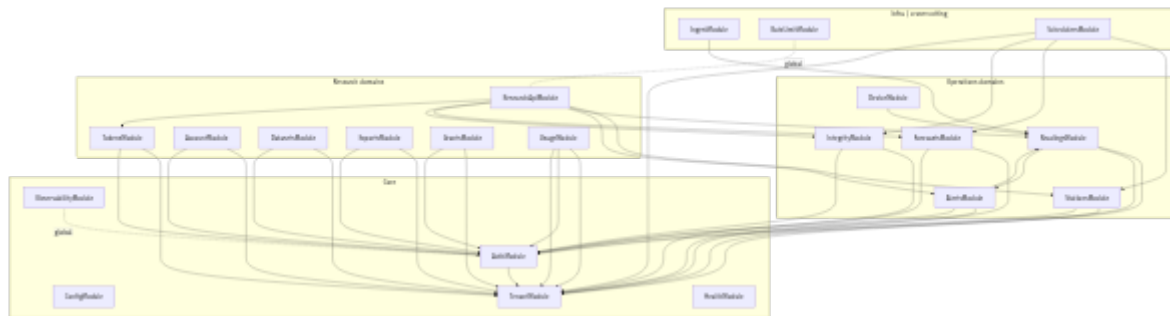
16. Rate-limit decision



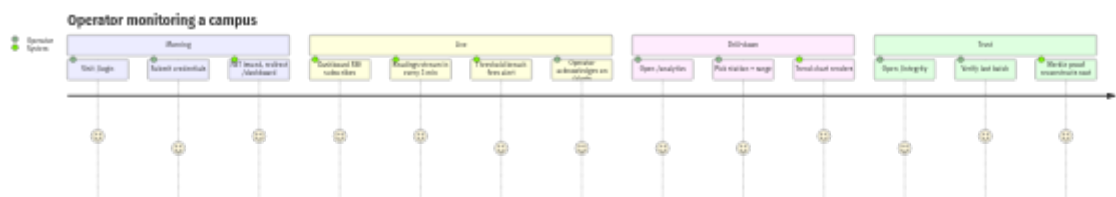
17. Observability — request → log line



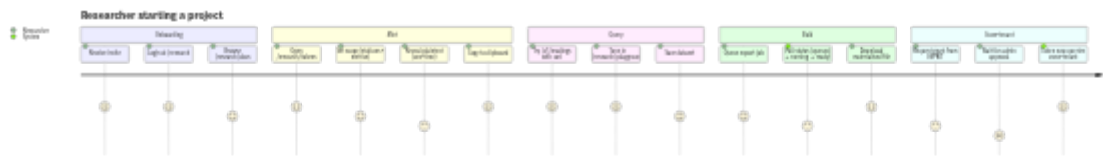
18. Module dependency graph



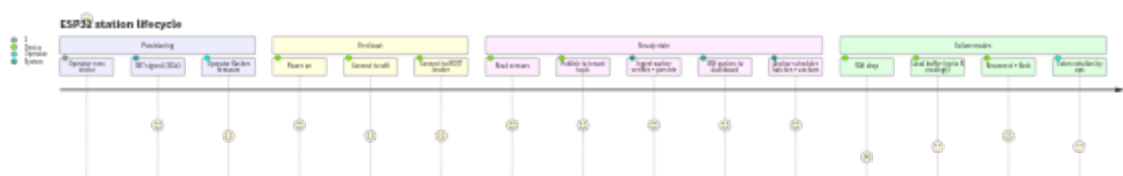
19. User journey — Operator daily routine



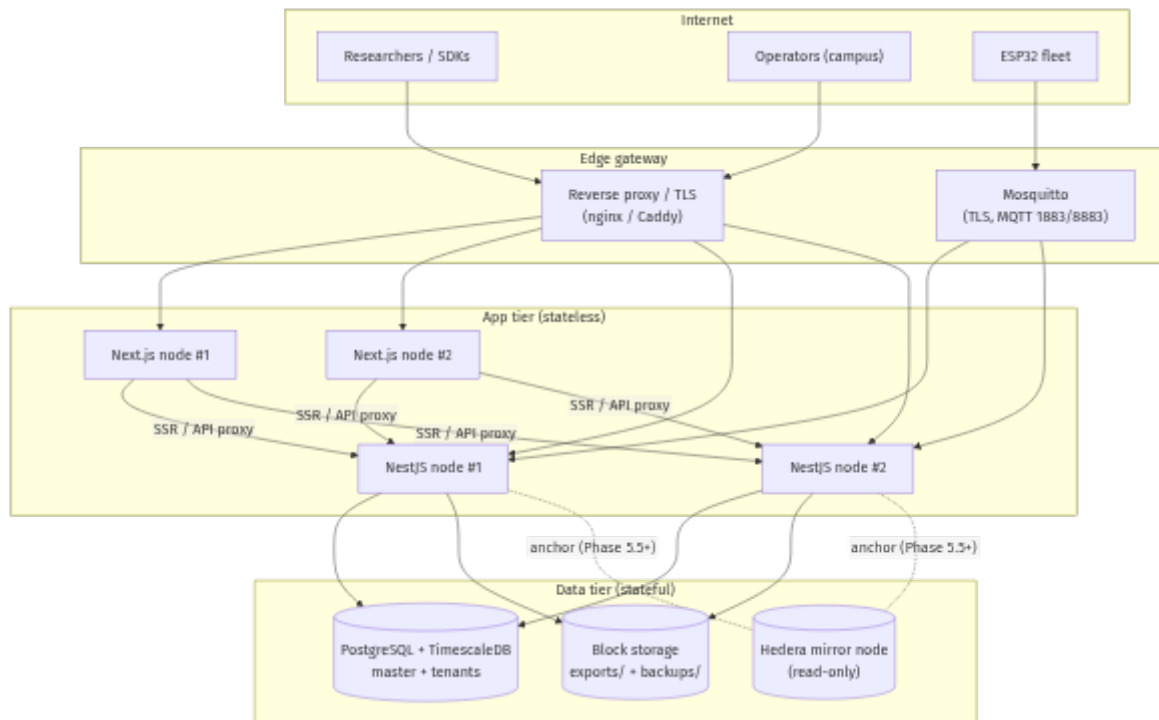
20. User journey — Researcher onboarding



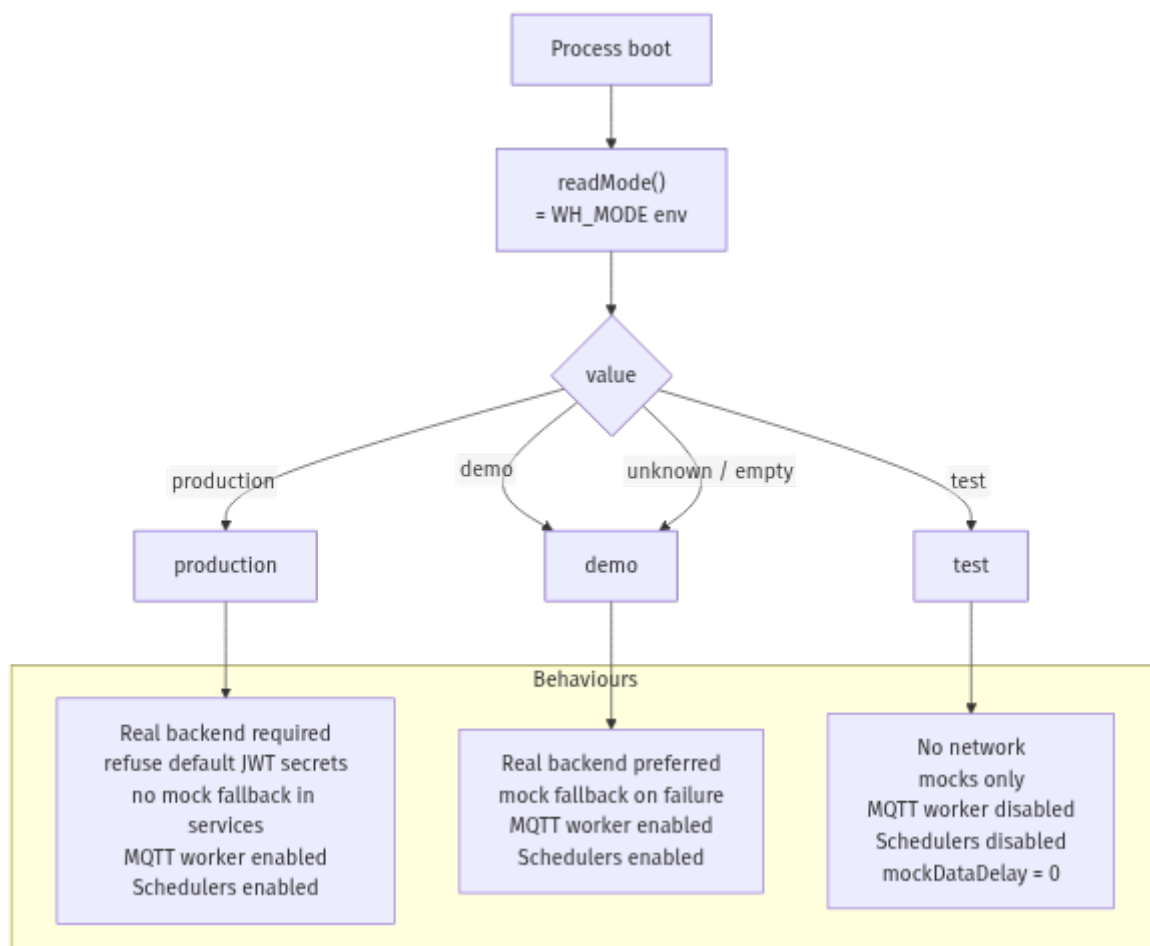
21. User journey — ESP32 device lifecycle



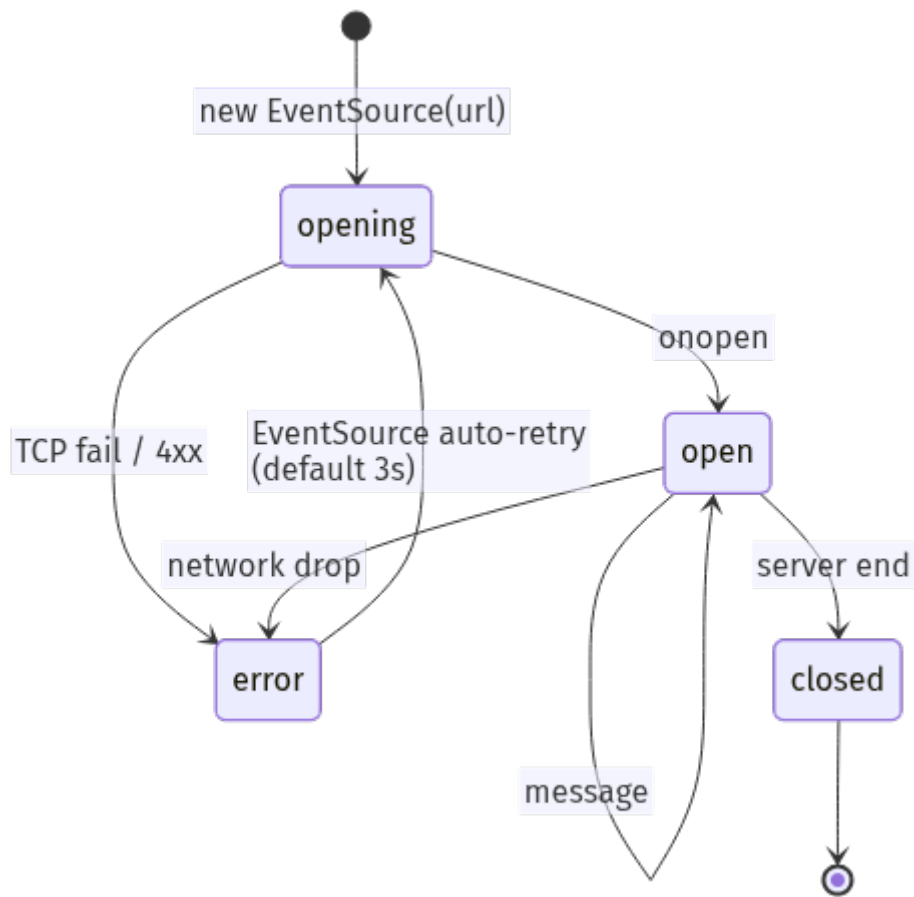
22. Deployment topology (target)



23. Configuration surface — `WH\_MODE` decision tree



24. Live SSE channel — connection states (browser side)



25. End-to-end timing budget — reading freshness

